ATVB IN FOCUS:
Immunomodulation of Cardiovascular Development, Growth, and Regeneration

Series Editor: Joy Lincoln

Immune Mechanisms of Heart Valve Development, Homeostasis, and Disease

Brittany A. Gonzalez, Katherine E. Yutzey

ABSTRACT: Valvular heart disease is among the most common cardiovascular conditions, usually presenting as aortic stenosis or mitral regurgitation. Clinically significant structural and functional anomalies can arise through congenital malformation, genetic variants, cardiovascular dysfunction, infectious disease sequelae, or chronic inflammation that can progress over time, necessitating surgical replacement or repair. The most common types of valve disease requiring surgical intervention are mitral valve prolapse, often with myxomatous changes due to congenital malformation or infectious disease, and aortic valve stenosis resulting from calcific aortic valve disease. Recent studies have identified critical contributions of immune cells to aortic and mitral valve pathophysiology. Moreover, preclinical studies demonstrate valve disease mechanisms related to innate or adaptive immune cell infiltration that could be targeted in human valvular heart disease using existing or emerging anti-inflammatory approaches. Here, we review recent findings of immune cell contributions and molecular mechanisms of myxomatous valve disease, rheumatic heart disease, and calcific aortic valve disease with potential therapeutic applications.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: congenital abnormalities ■ bicuspid aortic valve disease ■ heart valves ■ macrophages ■ rheumatic heart disease

Valvular heart disease (VHD) is among the most common cardiovascular conditions, with a prevalence of 2% to 4% in all adults in the United States, which increases to >10% in older individuals.[1–3] Clinically significant structural and functional anomalies are most common in left-sided aortic and mitral valves but can also affect pulmonary and tricuspid valves on the right side of the heart. The most common types of valve disease are mitral valve prolapse (MVP), often with myxomatous valve disease (MVD) due to congenital malformation or infectious disease, and aortic valve stenosis resulting from calcific aortic valve disease (CAVD; Figure 1). Subsequent valvular stenosis, regurgitation, or insufficiency can lead to reduced cardiac output, and potentially heart failure, if left untreated. The current standard of care for mitral or aortic valve disease is surgical or transcatheter-based repair or replacement, and there are currently limited medical therapies available. Recent preclinical studies have identified alterations in immune cells and inflammatory mechanisms as potential therapeutic targets in VHD.

Valve disease can arise through congenital malformation, heritable genetic variants, cardiovascular dysfunction, infectious diseases, or chronic inflammation that can progress over time, necessitating surgical replacement or repair.[4] Congenital valve malformations, such as a bicuspid aortic valve or MVP, can lead to accelerated valve dysfunction and disease, often manifesting in adulthood. Genetic conditions, such as Marfan syndrome (MFS), related to fibrillin-1 (Fbn1) variants, and other connective tissue disorders, such as Ehlers-Danlos syndrome, include heart valve pathology, such as myxomatous structural changes and MVP.[5–9] Rheumatic valve disease, characterized by valvulitis and tissue damage, arises as an autoimmune sequela of acute rheumatic fever due to *Streptococcus pyogenes* infection, which is preventable by antibiotic treatment, but has increased prevalence in low-resource settings.[10,11] Molecular and cellular studies

Explore the full series at www.ahajournals.org/atvb/atvb-focus; prior articles are listed in the endnotes.

Correspondence to: Katherine E. Yutzey, PhD, Division of Molecular Cardiovascular Biology, Cincinnati Children's Hospital Medical Center, ML7020, 240 Albert Sabin Way, Cincinnati, OH 45229. Email katherine.yutzey@cchmc.org

Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

CAVD	calcific aortic valve disease
CCL2	CC motif chemokine ligand 2
CCR2	CC motif chemokine receptor 2
DC	dendritic cell
ECM	extracellular matrix
IL	interleukin
IRAK-M	interleukin-37/interleukin-1 receptor-associated kinase M
MFS	Marfan syndrome
MHCII	major histocompatibility complex class II
MVD	myxomatous valve disease
MVP	mitral valve prolapse
P	postnatal
PPAR	peroxisome proliferator-activated receptor
RHD	rheumatic heart disease
SIRP	signal regulatory protein
TCR	T-cell receptor
TNF	tumor necrosis factor
VHD	valvular heart disease

Highlights

- Valvular heart diseases are among the most common cardiovascular conditions, usually presenting as aortic stenosis or mitral regurgitation.
- Recent studies have identified critical contributions of innate and adaptive immune cells to aortic and mitral valve pathophysiology.
- Preclinical studies support targeting of immune cell infiltration in valve disease using existing or emerging anti-inflammatory approaches.
- Targeting immune cells and related mechanisms that contribute to these valvular diseases represents a new therapeutic approach.

of VHD progression have led to new insights into potential therapeutic targets that might prevent the progression of calcific or myxomatous disease processes.[4]

Valve inflammation, consisting of both innate and adaptive immune cell infiltration, has been identified in progressive valve disease with developmental or genetic origins (MFS and bicuspid aortic valve), as well as with rheumatic heart disease (RHD), or with infective endocarditis that results from systemic bacterial infection.[2,12–18] Valve infiltrating and activated immune cells include myeloid lineage macrophages, dendritic cells (DCs), and mast cells, as well as T-cell populations. In a mouse model of MFS, genetic or pharmacological targeting of infiltrating CCR2 (CC motif chemokine receptor 2)+ myeloid lineage cells prevents mitral valve myxomatous degeneration.[2,12] Targeting inflammatory cytokines also may be an effective strategy, as was demonstrated for autoimmune cardiac valve inflammation in mice.[19] Thus, inflammatory processes represent attractive therapeutic targets that could take advantage of existing or emerging anti-inflammatory approaches. Here, we review recent findings of immune cell activation and contributions to VHD processes with potential therapeutic applications.

IMMUNE CELL LINEAGES IN CARDIOVASCULAR DEVELOPMENT AND HOMEOSTASIS

Immune cells are essential for maintaining cardiovascular health and integrity with functions in tissue repair, maintenance, and immune surveillance without triggering an inflammatory state.[20] Although immune cells have been studied extensively in myocardial homeostasis and injury, less is known about the contributions of immune cells to heart valve development and function. Recent studies in mice have identified diverse macrophage populations present in developing valves with dynamic changes in the postnatal period of valve remodeling.[21–24] During embryogenesis, immune cells present in the developing organs, including the heart, arise from multiple anatomic sites, including the yolk sac, aorta-gonad-mesonephros, and fetal liver.[14,25] While endocardial hematopoiesis has been implicated as a source of macrophages involved in prenatal valve morphogenesis,[21] extensive cell lineage studies demonstrate that immune and hematopoietic cells of the developing valves arise from multipotential progenitors in the yolk sac, aorta-gonad-mesonephros, and fetal liver, but not from endocardial endothelial cells.[25,26] Although diverse origins and types of myeloid lineage cells have been described, their functions in the prenatal heart valves are less clear.

Immune cells have been identified in murine aortic and mitral heart valves after birth at postnatal (P) day P1 and represent 5% to 10% of total heart valve cells.[22,24] Transcriptomic analysis of single cells from P7 and P30 aortic and mitral valves during valve leaflet remodeling uncovered distinct immune cell subsets in the valves, with similar relative proportions at each time point.[23] Distinct immune cell subpopulations were identified, including resident macrophages of embryonic and bone marrow origins, as well as T cells, DCs, and mast cells (Figure 1A and 1C).[23] During postnatal heart valve development, the proportion of yolk sac/embryonic-derived resident macrophages decreases, while the bone marrow–derived population increases by P30 and represents the majority of immune cells present, reflecting the developmental changes in myeloid lineage origins during the postnatal period.[14,22] More recently, transcriptomic analysis of single cells specifically in P1 murine aortic valves confirms that immune cell populations are primarily tissue resident macrophages, with little contribution of adaptive immune cells, such as DCs, mast cells, T cells, B cells, and natural killer cells.[24] These data further

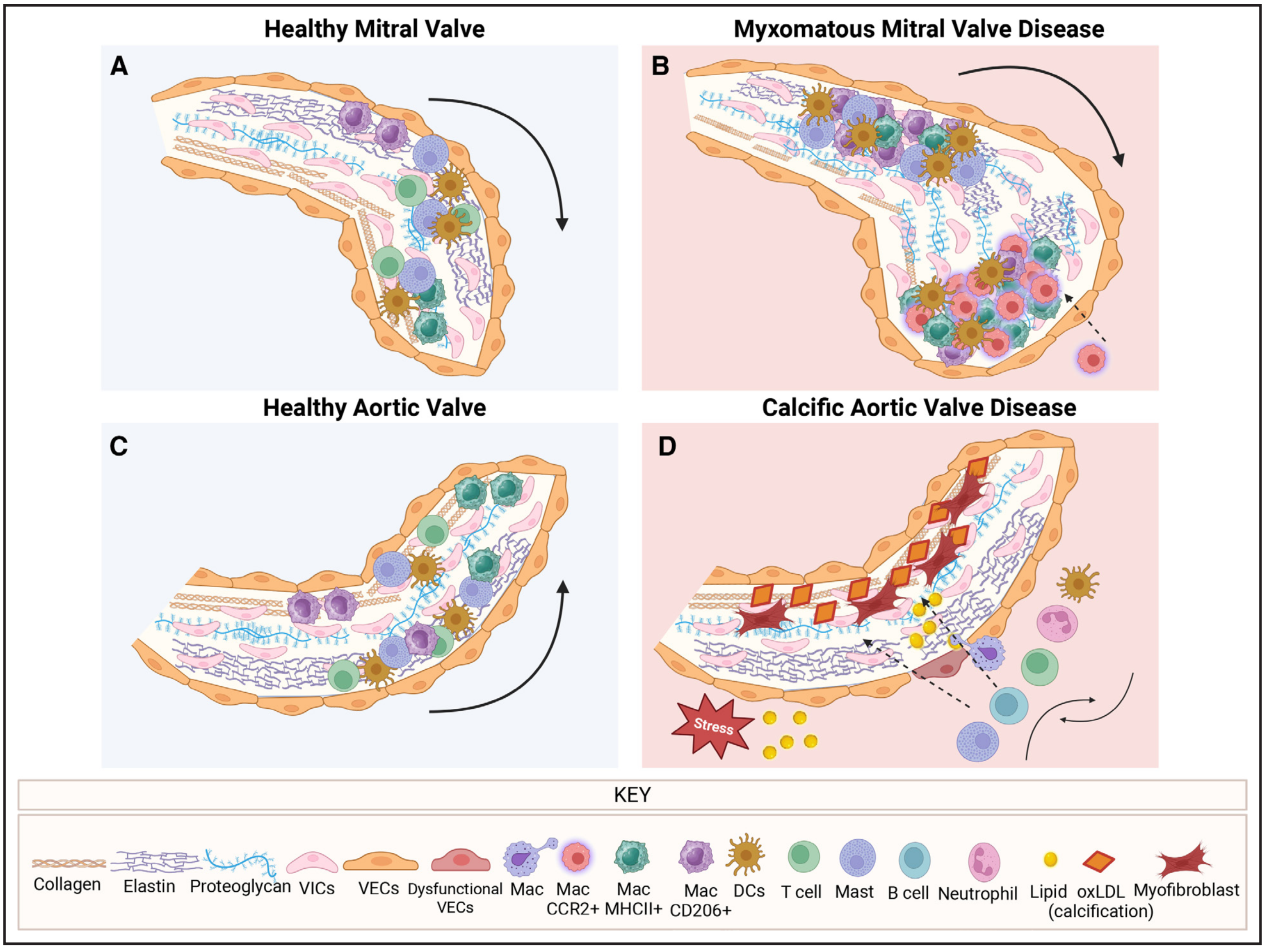

Figure 1. Extracellular matrix and inflammatory cells in mitral and aortic valve homeostasis and disease.
Valve cells and matrix proteins in healthy (**A**) mitral and (**C**) aortic valves, with the presence of collagen, elastin, and proteoglycans shown in each valve layer along with localized immune cells (macrophages, T cells, dendritic cells [DCs] and mast cells). **B**, In myxomatous mitral valve disease, the tissue becomes thicker with an increase in collagen, proteoglycans, and breakdown of elastin and infiltration of CCR2+ macrophages. **D**, In calcific aortic valve disease, lipids, LDL (low-density lipoprotein), and calcification occur along with myofibroblast activation and immune infiltration of macrophages, T cells, DCs, mast cells, and neutrophils. CCR2 indicates C-C motif chemokine receptor 2; Mac, macrophages; MHCII, major histocompatibility complex class II; oxLDL, oxidized low-density lipoprotein; VECs, valvular endothelial cells; and VICs, valvular interstitial cells. Created in BioRender. Gonzalez, BA. (2026) https://BioRender.com/5euisxd

support dynamic contributions of immune cells in the valves in the weeks after birth.

Adult mouse valves include resident immune cells, which are predominantly macrophages of embryonic and myeloid lineages, localized in regions of biomechanical stress at the commissure and hinge points of the leaflets.[12,22,27] Single-cell RNA sequencing analyses further demonstrate the presence of immune cells, which are predominantly resident macrophages in juvenile-adult mouse valves, although DCs, T cells, and B cells are also detected.[14,15,28] In adult human valves, multiple lymphocyte subclusters, including diverse macrophages and T cells, have been identified.[29] Single-cell RNA sequencing of all 4 valves combined (5 aortic valves, 5 pulmonary valves, 5 tricuspid valves, and 3 mitral valves) isolated from nondiseased human cardiac valve leaflets from patients with end-stage heart failure undergoing heart transplantation identified 6 major cell types with the largest population being valvular interstitial cells, followed by myeloid cells, lymphocytes, valvular endothelial cells, mast cells, and myofibroblasts.[29] Therefore, human heart valves contain multiple types of immune cells with likely functions in valve homeostasis and disease.

IMMUNE CELL LINEAGES IN MVD

Structural changes can occur in the mitral valve that lead to MVP, which affects 2% to 3% of the general population and can lead to significant mitral regurgitation, with a predisposition to bacterial endocarditis, congestive heart failure, and even sudden death.[30] In a murine model of MVP, regurgitation of both the mitral and aortic valves is associated with myxomatous valve progression at 2 and 6 months of age.[31] MVD is characterized at the histological level by accumulation of proteoglycans and fragmentation of collagen and elastic fibers (Figure 1B).

These ECM (extracellular matrix) changes can be caused by congenital malformations or genetic etiologies, cardiovascular dysfunction with progressive heart failure, autoimmune disease, or infectious endocarditis.[2,19,32,33] Moreover, mitral valves with regurgitation commonly exhibit features of MVD and can manifest due to intrinsic structural deficiencies in the leaflets or as a result of abnormal ventricular deformation and increased chamber size with advanced heart disease.[2,32]

Increased numbers of macrophages and other immune cells are seen in MVD, suggesting inflammation as a potential driver of the disease (Figure 1B).[2,12,34] CD45+ hematopoietic lineage cells, characterized as mostly macrophages with smaller numbers of DCs, are present in human, sheep, and murine diseased valves with myxomatous degeneration related to noninfective causes.[2,22,23,27,33,34] Increased inflammatory cells consisting of macrophages and T cells have been observed in human myxomatous mitral valves, along with upregulation of multiple cytokines, chemokines, and chemokine receptors, in both human and mouse diseased leaflets.[23] Furthermore, analysis of explanted human MVP tissue showed significantly increased lymphocyte and neutrophil numbers compared with controls.[35] Thus, there is an underlying inflammatory aspect of MVP in humans and animal models, but the role of inflammation in the pathogenesis of MVP is still not clear. The presence of increased immune cells in congenital or noninfectious MVD suggests that a sterile inflammatory response may be a common feature of valve disease from a variety of causes.

A mouse model for MVD associated with MFS, carrying a targeted missense C1041G mutation in the Fbn1 coding sequence, has been used to determine the origins and mechanisms of mitral valve pathology.[36,37] Gene expression profiling of *Fbn1*$^{C1041G/+}$ valves, compared with normal controls, confirmed inflammation and immune cell recruitment as increased in MFS valve disease in mice (Figure 1B).[2] Similarly, leukocytes and increased macrophages were also observed in gene edited pig MFS mitral valves, as well as human explanted myxomatous mitral valves. In the month after the birth of MFS mice, the heart valve leaflets exhibit abnormal ECM remodeling and structural maturation, but indicators of inflammation are not significantly increased.[2,14] Subsequently, in young adult MFS mice, infiltrating myeloid lineage cells are increased relative to endogenous resident immune cells normally present in healthy heart valves.[15,22] Notably, expression of CCL2 (CC motif chemokine ligand 2), a ligand for CCR2, is induced in resident valve interstitial cells in proximity to CCR2+ macrophages, and immunogenic proteoglycan matrix changes are present in regions of the valve where CCR2+ macrophages (lacking resident marker CD206) are increased specifically in MFS diseased valves.[2,12,38] Similar colocalization of modified hyaluronan and increased numbers of macrophages were observed in human MVD.[23] Together, these studies demonstrated that ECM changes and induced expression of chemokines and cytokines occur before the recruitment of circulating CCR2+ macrophages during the early disease stage of MVD.

The potential contributions of infiltrating myeloid cells to MVD in mice were examined using genetic or pharmacological targeting of CCR2+ cells. CCR2 genetic knockout studies demonstrated a reduction in mitral valve inflammation and inflammatory proteases, as well as restoration of valve morphology and ECM organization in a mouse model of MFS.[2] Similarly, treatment with the CCR2 antagonist RS504393 prevented macrophage infiltration and pathological ECM remodeling (Figure 2A).[13] More importantly, treatment with RS504393 restored immune cell localization along with reduced myxomatous valve degeneration and valve leaflet thickness.[13] Thus, CCR2+ immune cells contribute to valve myxomatous disease, and targeting CCR2+ infiltrating macrophages represents a potentially effective therapeutic approach for preventing or reversing MVD in mice.

IMMUNE MECHANISMS OF RHEUMATIC VHD

Valve inflammation, indicated by innate or adaptive immune cell infiltration, has also been identified in progressive valve dysfunction associated with RHD. RHD results from acute rheumatic fever caused by Group A Streptococcal infection and a subsequent autoimmune response with detrimental cardiac effects, including mitral valve disease.[10,11,39] Although rare in the United States, RHD-associated valve disease is more common in developing countries, with 250 000 deaths worldwide each year.[40] RHD is characterized by valve injury or dysfunction, such as heart valve scarring and deformity, narrowing or leaking of the valves, thickened and stiff leaflets, and fused commissures, necessitating replacement.[40] The treatment for RHD is an antibiotic targeting the specific infective organism and valve replacement or repair if the heart valve is damaged.

Autoimmune valve inflammation in RHD is evident in upregulation of cell adhesion molecules that activate lymphocyte infiltration, including T cells, which can then produce cytokines, such as TNF (tumor necrosis factor) and IL (interleukin), involved in macrophage activation and T-cell invasion (Figure 2B). This autoimmune process is different from sterile inflammatory mechanisms of congenital myxomatous valve degeneration described above, yet both result in ECM and valve leaflet structural abnormalities. Immune cell infiltration and inflammation have also been implicated in RHD in a rat model. Similar to the mouse MFS model, CCR2+ macrophages are also pivotal for abnormal valvular remodeling of RHD. In a rat model of RHD, there was an increase in infiltrating inflammatory cells, in particular, CCR2+ macrophage infiltration, and fibrosis in the valves, which were mitigated with

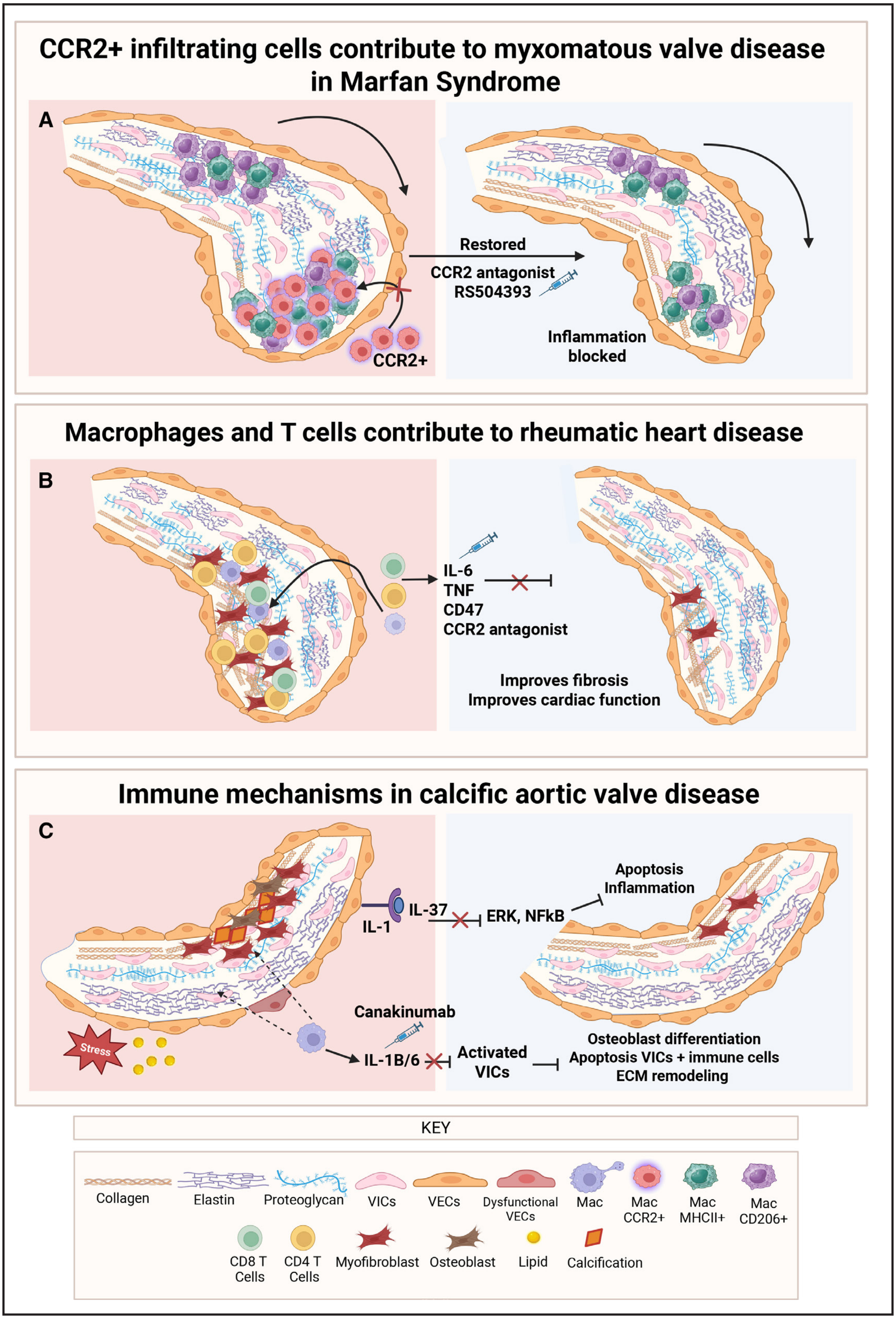

Figure 2. Therapeutic targeting of immune cells in heart valve disease.
Potential therapeutics for heart valve diseases of (**A**) myxomatous mitral valve disease (MVD), (**B**) rheumatic heart disease (RHD), and (**C**) calcific aortic valve disease (CAVD). **A**, CCR2+ macrophages infiltrate the valves with MVD. Blockade of CCR2+ infiltrating macrophages in MVD with CCR2 antagonist RS504393 alleviates the disease. **B**, Macrophages and T cells, including CD4 and CD8 T cells, contribute to RHD. Targeting IL6 (interleukin), cytokine TNF, or macrophages (CCR2+ and CD47) in RHD alleviates the disease. **C**, T cells and macrophages contribute to CAVD, but most interventions are focused on immune mechanisms by targeting interleukins (*Continued*)

a CCR2 antagonist treatment (Figure 2B).[41] CCR2+ and CCL2+ cells were also confirmed to be present in the valves of human patients with RHD. CCL2 was found to be expressed in leukocytes, endothelial cells, and fibroblasts, suggesting that recruitment of CCR2+ macrophages infiltrating from the bloodstream to the valve site occurs during disease progression.[41] Proteomic analysis of RHD and myxomatous valve degeneration in human patients has demonstrated distinct profiles of matrix and structural proteins that are differentially expressed between the 2 disease processes.[42] Thus, altered expression and tissue distribution of structural and matrix proteins are common features of MVD of different origins, but the disease processes are likely to be distinct depending on disease cause.

Progressive RHD, characterized by autoimmune-induced endocarditis with accompanying mitral valve disease, has been modeled in a well-studied K/BxN TCR (T-cell receptor) transgenic mouse line with autoimmune arthritis.[16] In these studies, mice with high-titer autoantibodies and arthritis had cardiac inflammation, specifically in the mitral and some aortic valves, starting at 3 weeks of age and progressively worsening over time.[16] The immune cells infiltrating the diseased mitral valves include macrophages, both tissue resident (F4/80 and CD11b) and bone marrow–derived (CD11b), as well as T cells (mostly CD4+ and some CD8+), with no detected B cells or neutrophils, which is similar to what is seen in rheumatic carditis.[16] Infiltrating mononuclear phagocytes were identified as key drivers of autoimmune MVD, and blockade of proinflammatory cytokines TNF and IL-6 led to improved cardiac outcomes in K/B.g7 mice (Figure 2B).[19] In a similar study, the interaction of CD47-expressing apoptotic cells with SIRP (signal regulatory protein) α on macrophages was tested to determine if they contribute to disease progression in K/B.g7 mice.[43] It was shown that anti-CD47 blockade prevented valvular carditis in K/B.g7 mice, as well as enhanced macrophage efferocytosis and reduced macrophage production of TNF and IL-6 (Figure 2B).[43] Thus, targeting of infiltrating macrophages and proinflammatory cytokines, including TNF and IL-6, may be an effective strategy to prevent autoimmune and RHD-associated heart valve disease.

IMMUNE CELL LINEAGES IN CAVD

Risk factors for aortic stenosis and progressive CAVD include older age, hypertension, atherosclerosis, diabetes, obesity, and kidney disease, as well as congenital malformations such as a bicuspid aortic valve.[44] Multiple indicators of inflammation are present in human calcified aortic valves, including chemokines/cytokines IL-1/IL-6 and TNF, as well as macrophage subtypes with similarities to foam cells of atherosclerosis.[45] However, CAVD differs from atherosclerotic disease in the lack of plaque-like luminal structures and in the presence of progressive calcification intrinsic to collagen-rich regions of the aortic valve cusps and hinge point regions.[46] Moreover, in contrast to MVD, macrophages with clear proinflammatory and foam cell-like properties are present in calcified aortic valves.[47] In addition, diseased aortic valves express indicators of oxidative stress and cell death that can lead to an inflammatory immune response, ultimately resulting in tissue calcification/mineralization that is not observed in myxomatous mitral valve leaflets.

Multiomics analysis of human CAVD shows the presence of multiple subtypes of macrophages, including foam cell-like and inflammatory subtypes, as well as T cells, in diseased valves (Figure 1D).[45] Resident and infiltrating macrophage subpopulations are present in diseased aortic valves and can differ with severity, but their specific contributions to disease progression have not been determined. In general, proinflammatory macrophages present in CAVD share myeloid origins and gene expression changes with macrophages present in atherosclerotic lesions, but the final pathology of arteries or valve cusps in terms of lesion formation or calcification is not the same.[17] Mechanical stress and endothelial damage occurring specifically in aortic valve leaflets during the cardiac cycle are potentially more severe in congenitally malformed valves and have been thought to be inflammatory contributors to CAVD.[48]

Specific contributions of immune cells to aortic valve thickening and hyperlipidemia before leaflet calcification have been examined in *apoE*$^{-/-}$; *Ldlr*$^{-/-}$ mice maintained on a Western diet.[17] Single-cell analysis of aortic valves from these adult hyperlipidemic mice demonstrates increased proportions of monocyte-derived (MHCII [major histocompatibility complex class II]-hi) macrophages, DCs, and T cells compared with controls. The monocyte-derived macrophages are CCR2+, and genetic loss of CCR2 reduces myeloid infiltration in hyperlipidemic valves. Moreover, the increase in valve inflammation and lipid accumulation can be inhibited by PPAR (peroxisome proliferator-activated receptor) gamma activation, supporting metabolic regulation of valve inflammation and disease in the context of hyperlipidemia.[28] Additional chemokine/cytokine signaling, as well as juxtacrine Notch signaling, in aortic valve macrophage subpopulations and resident cells have been identified.[17,49] Similarly, T cells are localized with other inflammatory indicators in early calcific lesions of diseased, but not healthy, human aortic valves.[44] Valve endothelial cell changes consistent with injury and proinflammatory mechanisms, which promote immune cell

Figure 2 Continued. (IL-1 b/6, IL-1/IL-37) in CAVD to alleviate disease and block inflammation. CCR2 indicates C-C motif chemokine receptor 2; Mac, macrophages; MHCII, major histocompatibility complex class II; VECs, valvular endothelial cells; and VICs, valvular interstitial cells. Created in BioRender. Gonzalez, BA. (2026) https://BioRender.com/5k0ten5 infiltration and downstream inflammation, occur in CAVD and may be initiating events in disease.[47] Further studies are needed to fully understand the immune cell components and interactions with resident cell populations that contribute to inflammatory processes in CAVD.

IMMUNOMODULATORY THERAPIES FOR VHD

Many anti-inflammatory compounds or biologics currently US Food and Drug Administration (FDA)–approved or in clinical trials could be repurposed for VHD therapeutics. As described above, recent studies in mice have identified potentially important new therapeutic targets for myxomatous and rheumatic valve disease. CCR2 inhibition demonstrated to prevent myeloid infiltration in MVD in mice is yet to be tested in humans (Figure 2A), but various CCR2 inhibitors are being tested in clinical trials for atherosclerosis, cardiac fibrosis, and cancer.[50] Mast cells have been observed to be increased in human stenotic aortic valves,[51] as well as in a myxomatous mitral valve mouse model of MFS, but potential contributions and functions in valve pathogenesis have not yet been determined. Because mast cell stabilization and degranulation can be targeted by FDA-approved therapies commonly used in allergy and asthma, these cells are an attractive therapeutic target for further investigation.[13] Similarly, FDA-approved therapeutics are available that target TNF and IL-6 implicated in autoimmune heart valve disease progression in the RHD model K/B.g7 mice (Figure 2B).[19] The recent increased availability of anti-inflammatory therapeutics developed for treating a variety of diseases will likely facilitate the translation of these animal findings to the treatment of clinically significant heart valve disease in the human population.

Other anti-inflammatory approaches specifically directed toward CAVD responses are needed. CAVD has some similarities to atherosclerosis, but, in large clinical trials (SALTIRE [Scottish Aortic Stenosis and Lipid Lowering Trial] and SEAS [Simvastatin and Ezetimibe in Aortic Stenosis]), anticholesterol statin therapies do not improve aortic valve function or stenosis outcomes.[52,53] An alternative approach targeting IL-1β was used in the CANTOS trial (Canakinumab Anti-Inflammation Thrombosis Outcomes Study) that showed reduced recurrent cardiac events in patients with previous myocardial infarction and cardiac damage who received canakinumab anti-inflammatory therapy (Figure 2C).[54] Because IL-1β is elevated in CAVD and studies in mice support reduced severity of aortic stenosis with loss of IL-6 receptor or genetic deficiency of IL-1β, canakinumab therapy might be beneficial for the treatment of CAVD (Figure 2C).[47] However, aortic valve parameters were not evaluated in the CANTOS trial, which was focused on coronary artery disease outcomes, and future studies focused on aortic stenosis and valve calcification are needed in support of this approach.[54]

The incidence of MVD is prevalent in some breeds of dogs, such as King Charles Spaniels and Golden retrievers, as well as more generally in small dogs, often leading to heart failure.[55] Currently, treatment is directed at maintaining cardiac output and preventing heart failure, but the high prevalence of valve disease in some breeds could be useful in developing clinical trials specifically targeted to MVD pathogenesis, including inflammation.[56] Preclinical studies are more advanced in mice, including targeting specific populations of immune cells in valve disease progression. As an example, genetic or pharmacological inhibition of CCR2 receptors, as described above, prevents infiltration of bone marrow–derived myeloid lineage cells, including macrophages, and prevents myxomatous disease in a mouse model of MFS (Figure 2A).[12,13] Multiple inflammatory targets have been identified in mouse CAVD, among which is an IRAK-M (IL-37/IL-1 receptor-associated kinase M) anti-inflammatory pathway (Figure 2C).[57] These preclinical studies show promise in the use of anti-inflammatory approaches and immune cell targeting in the prevention or regression of mitral and aortic valve disease.

PERSPECTIVES AND CLINICAL APPLICATIONS

Recent studies have demonstrated the presence of various immune cell types, including bone marrow–derived macrophages, DCs, T cells, and mast cells, in diseased heart valves from mice, pigs, and humans. Limited information is available on the specific pathogenic functions of these immune cells in valve pathogenesis, but preliminary studies in mouse models support anti-inflammatory approaches in the prevention and potential reversal of disease. Specific intervention in targeting of bone marrow–derived CCR2+ lineage cells or blockade of IL/receptor interactions currently in clinical trials could be beneficial in the treatment of VHD. Because the current standard treatment for valve disease is surgical or transcatheter-based replacement or repair, targeted anti-inflammatory pharmacological or biologic treatments would be an important advance in meeting the medical needs of this patient population.

ARTICLE INFORMATION

Received September 11, 2025; accepted May 4, 2026.

Affiliation

Division of Molecular Cardiovascular Biology, Cincinnati Children's Hospital Medical Center, The Heart Institute, University of Cincinnati College of Medicine, OH.

Acknowledgments

The graphic abstract was Created in BioRender. Gonzalez, BA. (2026) https://BioRender.com/546ug1j.

Author Contributions

B.A. Gonzalez and K.E. Yutzey were involved in drafting and editing. B.A. Gonzalez was involved in figure production.

Sources of Funding

Support for this work was provided by the National Institutes of Health (NIH) grants R01 HL143881 and R01 HL142217 to K.E. Yutzey, and the American Heart Association Postdoctoral Fellowship grant 23POST1019132 and NIH grant 1K99 HL179261-01 to B.A. Gonzalez.

Disclosures

None.

ATVB in Focus: Immunomodulation of Cardiovascular Development, Growth, and Regeneration. Series Editor: Joy Lincoln

Cardio-Hematopoietic Axis in Cardiac Injury and Repair: From Adaptation to Maladaptation
10.1161/ATVBAHA.125.323168
Immune Cell Regulation of Zebrafish Heart Regeneration
10.1161/ATVBAHA.125.323167
Immune Mechanisms of Heart Valve Development, Homeostasis, and Disease
10.1161/ATVBAHA.125.323160
Macrophages and the Survival of Cardiomyocytes After Myocardial Infarction
10.1161/ATVBAHA.125.323161

REFERENCES

1. Martin SS, Aday AW, Allen NB, Almarzooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Bansal N, Beaton AZ, et al; American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Committee. 2025 Heart disease and stroke statistics: a report of US and Global Data From the American Heart Association. *Circulation*. 2025;151:e41–e660. doi: 10.1161/CIR.0000000000001303
2. Kim AJ, Xu N, Yutzey KE. Macrophage lineages in heart valve development and disease. *Cardiovasc Res*. 2021;117:663–673. doi: 10.1093/cvr/cvaa062
3. Iung B, Vahanian A. Epidemiology of valvular heart disease in the adult. *Nat Rev Cardiol*. 2011;8:162–172. doi: 10.1038/nrcardio.2010.202
4. Small AM, Yutzey KE, Binstadt BA, Voigts KK, Bouatia-Naji N, Milan D, Aikawa E, Otto CM, St Hilaire C. Unraveling the mechanisms of valvular heart disease to identify medical therapy targets: a scientific statement from the American Heart Association. *Circulation*. 2024;150:e109–e128. doi: 10.1161/cir.0000000000001254
5. Weyman AE, Scherrer-Crosbie M. Marfan syndrome and mitral valve prolapse. *J Clin Invest*. 2004;114:1543–1546. doi: 10.1172/JCI23701
6. Kunkala MR, Schaff HV, Li Z, Volguina I, Dietz HC, LeMaire SA, Coselli JS, Connolly H. Mitral valve disease in patients with Marfan syndrome undergoing aortic root replacement. *Circulation*. 2013;128:S243–S247. doi: 10.1161/CIRCULATIONAHA.112.000113
7. Rashed ER, Ruiz Maya T, Black J, Fettig V, Kadian-Dodov D, Olin JW, Mehta L, Gelb BD, Kontorovich AR. Cardiovascular manifestations of hypermobile Ehlers-Danlos syndrome and hypermobility spectrum disorders. *Vasc Med*. 2022;27:283–289. doi: 10.1177/1358863X211067566
8. Acharya S, Kamat S, Shukla S, Giri A, Kumar S. Ehlers–Danlos syndrome type VI with mitral valve prolapse and regurgitation. *J Pract Cardiovasc Sci*. 2022;8:191–192. doi: 10.4103/jpcs.jpcs_40_22
9. Jaffe AS, Geltman EM, Rodey GE, Uitto J. Mitral valve prolapse: a consistent manifestation of type IV Ehlers-Danlos syndrome. The pathogenetic role of the abnormal production of type III collagen. *Circulation*. 1981;64:121–125. doi: 10.1161/01.cir.64.1.121
10. Lupieri A, Jha PK, Nizet V, Dutra WO, Nunes MCP, Levine RA, Aikawa E. Rheumatic heart valve disease: navigating the challenges of an overlooked autoimmune disorder. *Front Cardiovasc Med*. 2025;12:1537104. doi: 10.3389/fcvm.2025.1537104
11. Yeow S, Frost H, Porrello ER, Steer A, Voges HK. Valve biology and rheumatic heart disease pathogenesis. *Nat Cardiovasc Res*. 2026;5:204–217. doi: 10.1038/s44161-026-00792-9
12. Kim AJ, Xu N, Umeyama K, Hulin A, Ponny SR, Vagnozzi RJ, Green EA, Hanson P, McManus BM, Nagashima H, et al. Deficiency of circulating monocytes ameliorates the progression of myxomatous valve degeneration in Marfan syndrome. *Circulation*. 2020;141:132–146. doi: 10.1161/CIRCULATIONAHA.119.042391
13. Xu N, Yutzey KE. Therapeutic CCR2 blockade prevents inflammation and alleviates myxomatous valve disease in Marfan syndrome. *JACC Basic Transl Sci*. 2022;7:1143–1157. doi: 10.1016/j.jacbts.2022.06.001
14. Xu N, Gonzalez BA, Yutzey KE. Macrophage lineages in heart development and regeneration. *Curr Top Dev Biol*. 2024;156:1–17. doi: 10.1016/bs.ctdb.2024.01.004
15. Xu N, Alfieri CM, Yu Y, Guo M, Yutzey KE. Wnt signaling inhibition prevents postnatal inflammation and disease progression in mouse congenital myxomatous valve disease. *Arterioscler Thromb Vasc Biol*. 2024;44:1540–1554. doi: 10.1161/ATVBAHA.123.320388
16. Binstadt BA, Hebert JL, Ortiz-Lopez A, Bronson R, Benoist C, Mathis D. The same systemic autoimmune disease provokes arthritis and endocarditis via distinct mechanisms. *Proc Natl Acad Sci U S A*. 2009;106:16758–16763. doi: 10.1073/pnas.0909132106
17. Issa N, Blot G, Candellier A, Boudot C, Louvet L, Kamel S, Bennis Y, Hénaut L. Macrophages as key modulators of calcific aortic valve disease. *Front Cardiovasc Med*. 2025;12:1664067. doi: 10.3389/fcvm.2025.1664067
18. Tiveron MG, Pomerantzeff PMA, de Lourdes Higuchi M, Reis MM, de Jesus Pereira J, Kawakami JT, Ikegami RN, de Almeida Brandao CM, Jatene FB. Infectious agents is a risk factor for myxomatous mitral valve degeneration: a case control study. *BMC Infect Dis*. 2017;17:297. doi: 10.1186/s12879-017-2387-8
19. Meier LA, Auger JL, Engelson BJ, Cowan HM, Breed ER, Gonzalez-Torres MI, Boyer JD, Verma M, Marath A, Binstadt BA. CD301b/MGL2+ mononuclear phagocytes orchestrate autoimmune cardiac valve inflammation and fibrosis. *Circulation*. 2018;137:2478–2493. doi: 10.1161/CIRCULATIONAHA.117.033144
20. Wang X, Chen L, Wei J, Zheng H, Zhou N, Xu X, Deng X, Liu T, Zou Y. The immune system in cardiovascular diseases: from basic mechanisms to therapeutic implications. *Signal Transduct Target Ther*. 2025;10:166. doi: 10.1038/s41392-025-02220-z
21. Shigeta A, Huang V, Zuo J, Besada R, Nakashima Y, Lu Y, Ding Y, Pellegrini M, Kulkarni RP, Hsiai T, et al. Endocardially derived macrophages are essential for valvular remodeling. *Dev Cell*. 2019;48:617–630.e3. doi: 10.1016/j.devcel.2019.01.021
22. Hulin A, Anstine LJ, Kim AJ, Potter SJ, DeFalco T, Lincoln J, Yutzey KE. Macrophage transitions in heart valve development and myxomatous valve disease. *Arterioscler Thromb Vasc Biol*. 2018;38:636–644. doi: 10.1161/ATVBAHA.117.310667
23. Hulin A, Hortells L, Gomez-Stallons MV, O'Donnell A, Chetal K, Adam M, Lancellotti P, Oury C, Potter SS, Salomonis N, et al. Maturation of heart valve cell populations during postnatal remodeling. *Development*. 2019;146:dev173047. doi: 10.1242/dev.173047
24. Bluemn T, Kessler JR, Kim AJ, Drnevich J, Lincoln J. Murine aortic valve cell heterogeneity at birth. *Arterioscler Thromb Vasc Biol*. 2025;45:743–753. doi: 10.1161/ATVBAHA.124.322280
25. Liu K, Jin H, Tang M, Zhang S, Tian X, Zhang M, Han X, Liu X, Tang J, Pu W, et al. Lineage tracing clarifies the cellular origin of tissue-resident macrophages in the developing heart. *J Cell Biol*. 2022;221:e202108093. doi: 10.1083/jcb.202108093
26. Liu K, Jin H, Zhang S, Tang M, Meng X, Li Y, Pu W, Lui KO, Zhou B. Intercellular genetic tracing of cardiac endothelium in the developing heart. *Dev Cell*. 2023;58:1502–1512.e3. doi: 10.1016/j.devcel.2023.05.021
27. Hulin A, Moore V, James JM, Yutzey KE. Loss of Axin2 results in impaired heart valve maturation and subsequent myxomatous valve disease. *Cardiovasc Res*. 2017;113:40–51. doi: 10.1093/cvr/cvw229
28. Lee SH, Kim N, Kim M, Woo SH, Han I, Park J, Kim K, Park KS, Kim K, Shim D, et al. Single-cell transcriptomics reveal cellular diversity of aortic valve and the immunomodulation by PPARγ during hyperlipidemia. *Nat Commun*. 2022;13:5461. doi: 10.1038/s41467-022-33202-2
29. Shu S, Fu M, Chen X, Zhang N, Zhao R, Chang Y, Cui H, Liu Z, Wang X, Hua X, et al. Cellular landscapes of nondiseased human cardiac valves from end-stage heart failure-explanted heart. *Arterioscler Thromb Vasc Biol*. 2022;42:1429–1446. doi: 10.1161/ATVBAHA.122.318314
30. Delling FN, Vasan RS. Epidemiology and pathophysiology of mitral valve prolapse: new insights into disease progression, genetics, and molecular basis. *Circulation*. 2014;129:2158–2170. doi: 10.1161/CIRCULATIONAHA.113.006702
31. Gonzalez BA, Harmeyer SW, Song T, Sadayappan S, Yutzey KE. Dynamic changes in mitral valve extracellular matrix, tissue mechanics and function in a mouse model of Marfan syndrome. *Matrix Biol*. 2024;126:1–13. doi: 10.1016/j.matbio.2024.01.001
32. Levine RA, Hagége AA, Judge DP, Padala M, Dal-Bianco JP, Aikawa E, Beaudoin J, Bischoff J, Bouatia-Naji N, Bruneval P, et al; Leducq Mitral

Transatlantic Network. Mitral valve disease--morphology and mechanisms. *Nat Rev Cardiol*. 2015;12:689–710. doi: 10.1038/nrcardio.2015.161

33. Geirsson A, Singh M, Ali R, Abbas H, Li W, Sanchez JA, Hashim S, Tellides G. Modulation of transforming growth factor-β signaling and extracellular matrix production in myxomatous mitral valves by angiotensin II receptor blockers. *Circulation*. 2012;126:S189–S197. doi: 10.1161/CIRCULATIONAHA.111.082610
34. Sauls K, Toomer K, Williams K, Johnson AJ, Markwald RR, Hajdu Z, Norris RA. Increased infiltration of extra-cardiac cells in myxomatous valve disease. *J Cardiovasc Dev Dis*. 2015;2:200–213. doi: 10.3390/jcdd2030200
35. Yalim Z, Ersoy I. Evaluation of inflammation markers in mitral valve prolapse. *Arch Cardiol Mex*. 2022;92:181–188. doi: 10.24875/ACM.21000127
36. Judge DP, Biery NJ, Keene DR, Geubtner J, Myers L, Huso DL, Sakai LY, Dietz HC. Evidence for a critical contribution of haploinsufficiency in the complex pathogenesis of Marfan syndrome. *J Clin Invest*. 2004;114:172–181. doi: 10.1172/JCI20641
37. Ng CM, Cheng A, Myers LA, Martinez-Murillo F, Jie C, Bedja D, Gabrielson KL, Hausladen JM, Mecham RP, Judge DP, et al. TGF-beta-dependent pathogenesis of mitral valve prolapse in a mouse model of Marfan syndrome. *J Clin Invest*. 2004;114:1586–1592. doi: 10.1172/JCI22715
38. Petrey AC, de la Motte CA. Hyaluronan, a crucial regulator of inflammation. *Front Immunol*. 2014;5:101. doi: 10.3389/fimmu.2014.00101
39. Veinot JP. Pathology of inflammatory native valvular heart disease. *Cardiovasc Pathol*. 2006;15:243–251. doi: 10.1016/j.carpath.2006.04.007
40. Seckeler MD, Hoke TR. The worldwide epidemiology of acute rheumatic fever and rheumatic heart disease. *Clin Epidemiol*. 2011;3:67–84. doi: 10.2147/CLEP.S12977
41. Bai L, Li Y, Xue Y, Lu Z, Meng Z, Lu C, Huang F, Zeng Z. Inhibition of macrophage recruitment to heart valves mediated by the C-C chemokine receptor type 2 attenuates valvular inflammation induced by Group A Streptococcus in Lewis rats. *Front Biosci (Landmark Ed)*. 2024;29:303. doi: 10.31083/j.fbl2908303
42. Martins CO, Santos KS, Ferreira FM, Teixeira PC, Pomerantzeff PM, Brandão CM, Sampaio RO, Spina GS, Kalil J, Guilherme L, et al. Distinct mitral valve proteomic profiles in rheumatic heart disease and myxomatous degeneration. *Clin Med Insights Cardiol*. 2014;8:79–86. doi: 10.4137/CMC.S17622
43. Meier LA, Faragher JL, Osinski V, Auger JL, Voeller R, Marath A, Binstadt BA. CD47 promotes autoimmune valvular carditis by impairing macrophage efferocytosis and enhancing cytokine production. *J Immunol*. 2022;208:2643–2651. doi: 10.4049/jimmunol.2100903
44. Otto CM, Prendergast B. Aortic-valve stenosis--from patients at risk to severe valve obstruction. *N Engl J Med*. 2014;371:744–756. doi: 10.1056/NEJMra1313875
45. Villa-Roel N, Park C, Andueza A, Baek KI, Su A, Blaser MC, Leshnower BG, Yoganathan A, Aikawa E, Jo H. Side- and disease-dependent changes in human aortic valve cell population and transcriptomic heterogeneity determined by single-cell RNA sequencing. *Genes (Basel)*. 2024;15:1623. doi: 10.3390/genes15121623
46. Gomez-Stallons MV, Tretter JT, Hassel K, Gonzalez-Ramos O, Amofa D, Ollberding NJ, Mazur W, Choo JK, Smith JM, Kereiakes DJ, et al. Calcification and extracellular matrix dysregulation in human postmortem and surgical aortic valves. *Heart*. 2019;105:1616–1621. doi: 10.1136/heartjnl-2019-314879
47. Bartoli-Leonard F, Zimmer J, Aikawa E. Innate and adaptive immunity: the understudied driving force of heart valve disease. *Cardiovasc Res*. 2021;117:2506–2524. doi: 10.1093/cvr/cvab273
48. Dayawansa NH, Baratchi S, Peter K. Uncoupling the vicious cycle of mechanical stress and inflammation in calcific aortic valve disease. *Front Cardiovasc Med*. 2022;9:783543. doi: 10.3389/fcvm.2022.783543
49. Klauzen P, Basovich L, Shishkova D, Markova V, Malashicheva A. Macrophages in calcific aortic valve disease: paracrine and juxtacrine disease drivers. *Biomolecules*. 2024;14:1547. doi: 10.3390/biom14121547
50. Guo S, Zhang Q, Guo Y, Yin X, Zhang P, Mao T, Tian Z, Li X. The role and therapeutic targeting of the CCL2/CCR2 signaling axis in inflammatory and fibrotic diseases. *Front Immunol*. 2024;15:1497026. doi: 10.3389/fimmu.2024.1497026
51. Wypasek E, Natorska J, Grudzień G, Filip G, Sadowski J, Undas A. Mast cells in human stenotic aortic valves are associated with the severity of stenosis. *Inflammation*. 2013;36:449–456. doi: 10.1007/s10753-012-9565-z
52. Cowell SJ, Newby DE, Prescott RJ, Bloomfield P, Reid J, Northridge DB, Boon NA; Scottish Aortic Stenosis and Lipid Lowering Trial, Impact on Regression (SALTIRE) Investigators. A randomized trial of intensive lipid-lowering therapy in calcific aortic stenosis. *N Engl J Med*. 2005;352:2389–2397. doi: 10.1056/NEJMoa043876
53. Rossebø AB, Pedersen TR, Boman K, Brudi P, Chambers JB, Egstrup K, Gerdts E, Gohlke-Bärwolf C, Holme I, Kesäniemi YA, et al; SEAS Investigators. Intensive lipid lowering with simvastatin and ezetimibe in aortic stenosis. *N Engl J Med*. 2008;359:1343–1356. doi: 10.1056/NEJMoa0804602
54. Ridker PM, Everett BM, Thuren T, MacFadyen JG, Chang WH, Ballantyne C, Fonseca F, Nicolau J, Koenig W, Anker SD, et al; CANTOS Trial Group. Antiinflammatory therapy with canakinumab for atherosclerotic disease. *N Engl J Med*. 2017;377:1119–1131. doi: 10.1056/NEJMoa1707914
55. Markby G, Summers KM, MacRae VE, Del-Pozo J, Corcoran BM. Myxomatous degeneration of the canine mitral valve: from gross changes to molecular events. *J Comp Pathol*. 2017;156:371–383. doi: 10.1016/j.jcpa.2017.01.009
56. Markby GR, Summers KM, MacRae VE, Corcoran BM. Comparative transcriptomic profiling and gene expression for myxomatous mitral valve disease in the dog and human. *Vet Sci*. 2017;4:34. doi: 10.3390/vetsci4030034
57. Xian G, Huang R, Hu D, Xu M, Chen Y, Ren H, Xu D, Zeng Q. Interleukin-37 attenuates aortic valve lesions by inhibiting N6-methyladenosine-mediated interleukin-1 receptor-associated kinase M degradation. *Cardiovasc Res*. 2025;121:492–506. doi: 10.1093/cvr/cvaf012